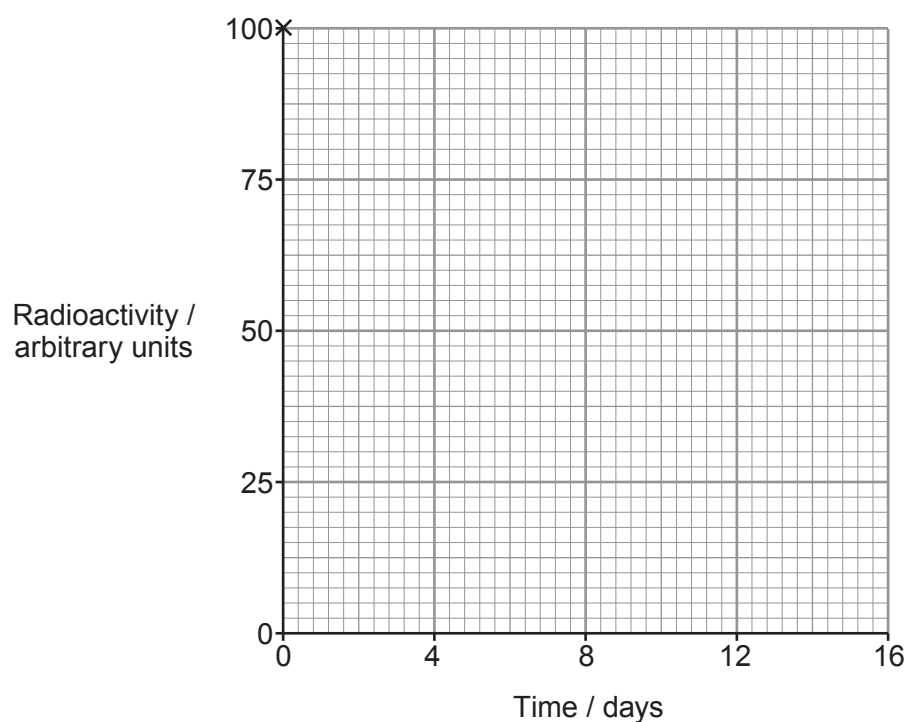


10. (a) Iodine-124 is an unstable radioactive isotope that decays by positron emission. It has a half-life of 4 days.

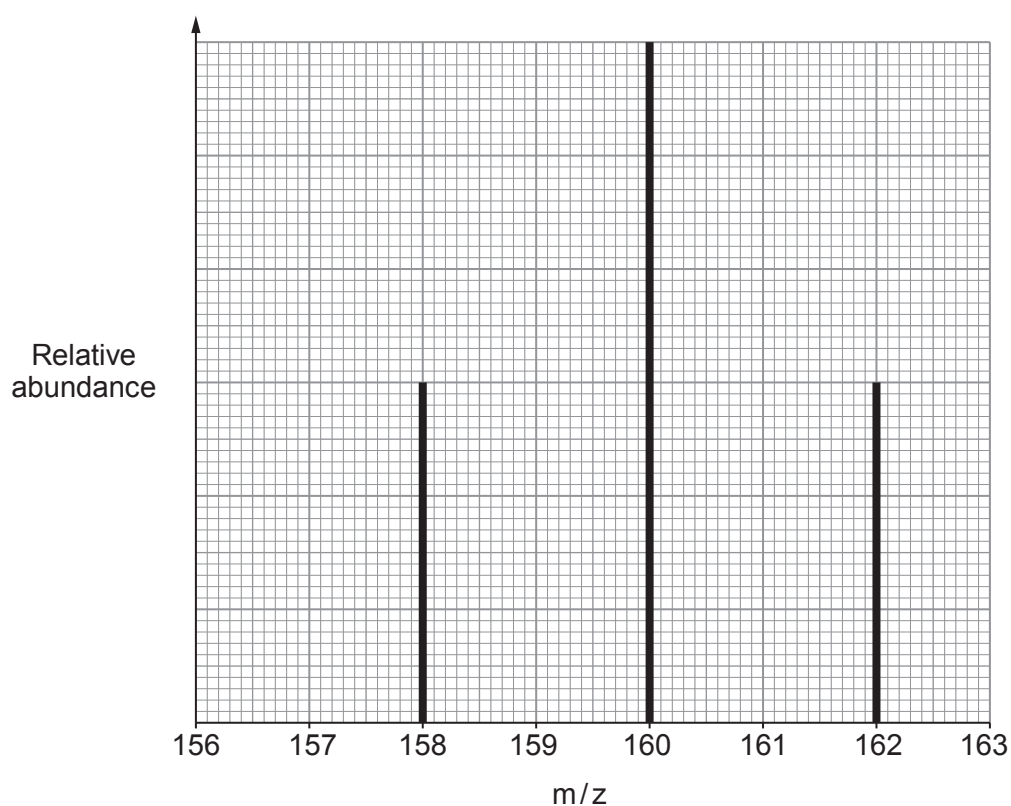
(i) Write an equation to show this decay. [2]

.....

- (ii) Complete the graph below to show how the radioactivity of a sample of iodine-124 would vary with time. [2]



(b) The molecular ion region of the mass spectrum of bromine, Br_2 , is shown below.



Explain the presence of the three peaks and why their height ratio is 1:2:1.

[3]

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.....

(c) A student was asked to find the pH of an acidic solution. She used a pH meter and found it to be 1.92.

(i) Calculate the hydrogen ion concentration of this solution.

[1]

$[\text{H}^+] = \dots\dots\dots \text{mol dm}^{-3}$



- (ii) Since the pH meter had not been used for a while, the teacher wanted to check the accuracy of the reading.

He asked the student to check this by titrating 25.0 cm^3 of the acidic solution against standardised barium hydroxide solution and calculating the hydrogen ion concentration using the following equation.



These were the results.

Volume of acidic solution	25.0 cm^3
Concentration of Ba(OH)_2	$5.02 \times 10^{-3}\text{ mol dm}^{-3}$
Titre	23.55 cm^3

Use these results and the equation to calculate the number of moles of hydrogen ions, H^+ , in 25.0 cm^3 of the acidic solution and hence the hydrogen ion concentration. [3]

$$[\text{H}^+] = \dots\dots\dots \text{mol dm}^{-3}$$

- (iii) The student said that the reading on the pH meter was incorrect. Do you agree? Justify your answer by comparing your answers to parts (i) and (ii). [1]

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